Monitoring Stack Setup Guide

Prometheus + Grafana + Node Exporter on Ubuntu Server 22.04 LTS

Overview

This guide documents the full setup of a production-style monitoring stack on a dedicated Ubuntu Server VM. It covers infrastructure provisioning, OS hardening, and deployment of Prometheus, Node Exporter, and Grafana — a standard open-source observability stack used in professional DevOps environments.

Stack

Tool	Role	Port
Prometheus	Metrics collection & storage	9090
Node Exporter	Hardware & OS metrics exporter	9100
Grafana	Metrics visualization & dashboards	3000

Architecture

```
[Target VMs] → Node Exporter (9100)
                        ↓
               Prometheus (9090) ← scrapes metrics
                        ↓
                Grafana (3000) ← queries & visualizes
```

Variables — Adapt Before Use

```
MONITORING_IP=<monitoring-vm-ip>      # e.g. 192.168.x.x
GATEWAY=<your-gateway-ip>             # e.g. 192.168.x.1
SUBNET=<your-subnet>                  # e.g. 192.168.x.0/24
SSH_KEY_PATH=<path-to-your-key>       # e.g. ~/.ssh/monitoring-vm
USER=<your-username>
```

Table of Contents

1. VM Provisioning
2. Ubuntu Server Installation
3. SSH Key Authentication
4. SSH Hardening
5. Firewall Configuration
6. System Updates
7. Prometheus Installation
8. Node Exporter Installation
9. Prometheus Configuration
10. Grafana Installation
11. Grafana Configuration
12. Verification & Access

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1. VM Provisioning

Recommended Specifications

Parameter	Recommended Value	Notes
CPU	1–2 cores	1 core sufficient for small labs
RAM	2–4 GB	2GB minimum for Prometheus + Grafana
Disk	20–50 GB	More if storing long-term metrics
Network	Static IP	Required for stable scraping targets
OS	Ubuntu Server 22.04 LTS	LTS recommended for stability

Proxmox-Specific Notes

- Set CPU type to `host` for best performance on single-node setups
- Use **VirtIO** network driver for best throughput
- Enable **iothread** on SCSI disk for better I/O performance
- Disable **High Availability** on single-node Proxmox

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2. Ubuntu Server Installation

Network Configuration (Static IP)

During installation, configure the network manually:

Field	Value
IPv4 Method	Manual
Subnet	`$SUBNET`
Address	`$MONITORING_IP`

Field	Value
Gateway	`$GATEWAY`
DNS	`1.1.1.1, 8.8.8.8`

Storage Layout

The default LVM layout is recommended:

Mount Point	Suggested Size	Type
`/boot`	2 GB	ext4
`/`	14–30 GB	ext4 (LVM)

SSH Configuration

During installation:

- **Install OpenSSH server** — enable
- **Allow password authentication** — enable temporarily *(disabled after key setup)*

Featured Snaps

Skip all snaps. Install Prometheus and Grafana manually via `apt` for version control and reliability.

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3. SSH Key Authentication

Generate SSH Key Pair

Linux/macOS:

```
ssh-keygen -t ed25519 -f ~/.ssh/monitoring-vm -C "monitoring-vm"
```

Windows PowerShell:

```
ssh-keygen -t ed25519 -f "$env:USERPROFILE\.ssh\monitoring-vm" -C
"monitoring-vm"
```

Copy Public Key to VM

Linux/macOS:

```
ssh-copy-id -i ~/.ssh/monitoring-vm.pub $USER@$MONITORING_IP
```

Windows PowerShell:

```
type $env:USERPROFILE\.ssh\monitoring-vm.pub | ssh $USER@$MONITORING_IP
"mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys &&
chmod 600 ~/.ssh/authorized_keys"
```

Verify Key Installation

```
cat ~/.ssh/authorized_keys
```

Test Key-Based Login

Linux/macOS:

```
ssh -i ~/.ssh/monitoring-vm $USER@$MONITORING_IP
```

Windows PowerShell:

```
ssh -i "$env:USERPROFILE\.ssh\monitoring-vm" $USER@$MONITORING_IP
```

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4. SSH Hardening

> *Run on the VM after confirming key login works.*

Apply Hardening

```
sudo sed -i 's/#\?PasswordAuthentication.*/PasswordAuthentication no/'
/etc/ssh/sshd_config
sudo sed -i 's/#\?PubkeyAuthentication.*/PubkeyAuthentication yes/'
/etc/ssh/sshd_config
sudo sed -i 's/#\?PermitRootLogin.*/PermitRootLogin no/'
/etc/ssh/sshd_config
```

Verify Configuration

```
grep -E "PasswordAuthentication|PubkeyAuthentication|PermitRootLogin"
/etc/ssh/sshd_config
```

Expected output:

```
PasswordAuthentication no
PubkeyAuthentication yes
PermitRootLogin no
```

Restart SSH Service

```
sudo systemctl restart sshd
```

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5. Firewall Configuration

Set Default Policies

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
```

Allow Required Ports

```
sudo ufw allow 22/tcp      # SSH
sudo ufw allow 9090/tcp    # Prometheus
sudo ufw allow 3000/tcp    # Grafana
```

> *In production, restrict Prometheus and Grafana access to trusted IPs only:*
>
> ```
> sudo ufw allow from <trusted-ip> to any port 9090
> sudo ufw allow from <trusted-ip> to any port 3000
> ```

Enable & Verify

```
sudo ufw enable
sudo ufw status
```

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6. System Updates

```
sudo apt update && sudo apt upgrade -y
sudo apt autoremove -y
```

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7. Prometheus Installation

Install

```
sudo apt install prometheus -y
```

Enable & Start

```
sudo systemctl enable prometheus
sudo systemctl start prometheus
```

Verify

```
sudo systemctl status prometheus
```

Access UI: `http://$MONITORING_IP:9090`

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8. Node Exporter Installation

Node Exporter exposes system-level metrics (CPU, RAM, disk, network) to Prometheus.

Install

```
sudo apt install prometheus-node-exporter -y
```

Enable & Start

```
sudo systemctl enable prometheus-node-exporter
sudo systemctl start prometheus-node-exporter
```

Verify

```
sudo systemctl status prometheus-node-exporter
```

Metrics available at: `http://$MONITORING_IP:9100/metrics`

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9. Prometheus Configuration

Edit Prometheus Config

```
sudo nano /etc/prometheus/prometheus.yml
```

Add Scrape Targets

```
scrape_configs:

  # Prometheus self-monitoring (default)
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  # Local Node Exporter
  - job_name: 'node'
    static_configs:
      - targets: ['localhost:9100']

  # Example: scrape a remote target
  # - job_name: 'app-vm'
  #   static_configs:
  #     - targets: ['<app-vm-ip>:9100']
```

Apply Changes

```
sudo systemctl restart prometheus
```

Verify Targets

1. Go to `http://$MONITORING_IP:9090`
2. Navigate to **Status** → **Targets**
3. All jobs should show status **UP** ☒

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10. Grafana Installation

Add Repository (Modern Method)

```
wget -q -O /tmp/grafana.gpg https://packages.grafana.com/gpg.key
sudo gpg --dearmor -o /usr/share/keyrings/grafana.gpg /tmp/grafana.gpg
echo "deb [signed-by=/usr/share/keyrings/grafana.gpg]
https://packages.grafana.com/oss/deb stable main" \
  | sudo tee /etc/apt/sources.list.d/grafana.list
```

Install

```
sudo apt update && sudo apt install grafana -y
```

Enable & Start

```
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
```

Verify

```
sudo systemctl status grafana-server
```

> *On first start, Grafana runs database migrations. Wait ~60 seconds before accessing the UI.*

Access UI: `http://$MONITORING_IP:3000`

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11. Grafana Configuration

First Login

- URL: `http://$MONITORING_IP:3000`
- Default credentials: `admin` / `admin`
- **Change password immediately on first login**

Add Prometheus Data Source

1. **Connections** → **Data sources** → **Add data source**
2. Select **Prometheus**
3. URL: `http://$MONITORING_IP:9090`
4. Authentication: None (internal network)
5. Click **Save & Test** → expect green

Import Pre-Built Dashboard

1. **Dashboards** → **New** → **Import**
2. Enter ID: `1860` *(Node Exporter Full — community dashboard)*
3. Click **Load**
4. Select **Prometheus** as data source
5. Click **Import**

> *Other useful dashboard IDs:*
>
> - `3662` *— Prometheus 2.0 Stats*
> - `7362` *— Node Exporter with Full Metrics*

> - `11074` — *Node Exporter for Prometheus*

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12. Verification & Access

Services Checklist

Service	Port	Check Command
SSH	22	`sudo systemctl status sshd`
Prometheus	9090	`sudo systemctl status prometheus`
Node Exporter	9100	`sudo systemctl status prometheus-node-exporter`
Grafana	3000	`sudo systemctl status grafana-server`

All Services at Once

```
for svc in sshd prometheus prometheus-node-exporter grafana-server; do
  echo "=== $svc ===" && sudo systemctl is-active $svc
done
```

Optional: SSH Config Alias

Add to `~/.ssh/config` on your local machine:

```
Host monitoring-vm
    HostName <MONITORING_IP>
    User <USER>
    IdentityFile <SSH_KEY_PATH>
    StrictHostKeyChecking no
```

Usage:

```
ssh monitoring-vm
```

Security Checklist

- [ ] SSH password authentication disabled
- [ ] Root SSH login disabled
- [ ] SSH key authentication enabled and tested
- [ ] UFW firewall active with only required ports open
- [ ] System packages fully updated
- [ ] Grafana default password changed
- [ ] Prometheus not exposed to the internet
- [ ] Grafana not exposed to the internet
- [ ] Node Exporter port (9100) restricted to internal network only

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Troubleshooting

Grafana Won't Load

```
sudo systemctl status grafana-server
sudo journalctl -u grafana-server -n 50
```

Wait 60 seconds after start — Grafana runs DB migrations on first boot.

Prometheus Targets Show DOWN

```
# Check Node Exporter is running
sudo systemctl status prometheus-node-exporter

# Confirm port is open
ss -tlnp | grep 9100
```

```
# Check firewall
sudo ufw status
```

SSH Access Denied After Hardening

```
# From Proxmox console (direct VM access)
sudo nano /etc/ssh/sshd_config
# Re-enable PasswordAuthentication temporarily
sudo systemctl restart sshd
```

Permission Issues on `.ssh` Directory

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
chmod 600 ~/.ssh/id_ed25519
chmod 644 ~/.ssh/id_ed25519.pub
```